

4.1.2. Newton's 2nd Law {Rigid Body}

* Rotational Inertia {Mass Moment of Inertia}

* Newton's 2nd Law of Motion in FAR

* Newton's 2nd Law of Motion in GPM.

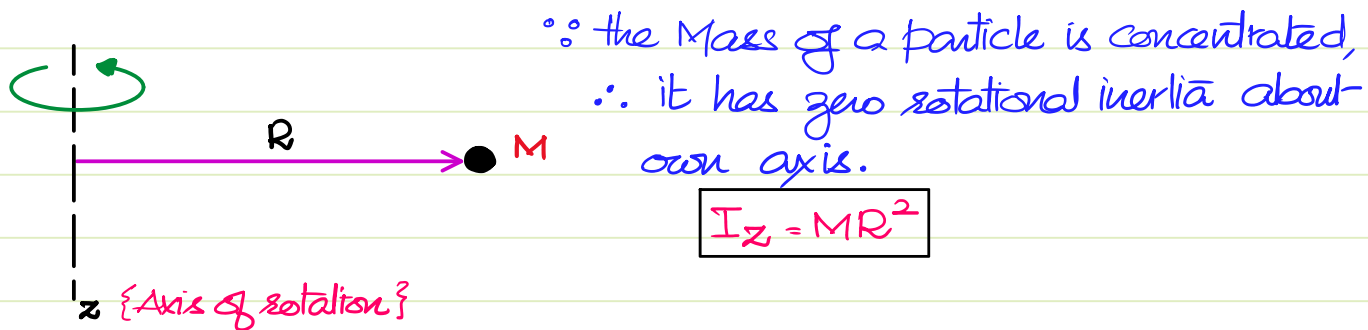
* Rolling friction

* Vehicle friction

* Rotational Inertia {Mass M.O.I.}: During rotation the resistance offered by a body against acceleration is due to Mass & also the distribution of Mass w.r.t. the axis of rotation. This resistance is known as **Mass Moment of Inertia**.

$I = f\{\text{Mass, Distribution of Mass w.r.t. axis of rotation}\}$

A. Rotational Inertia of Particle :

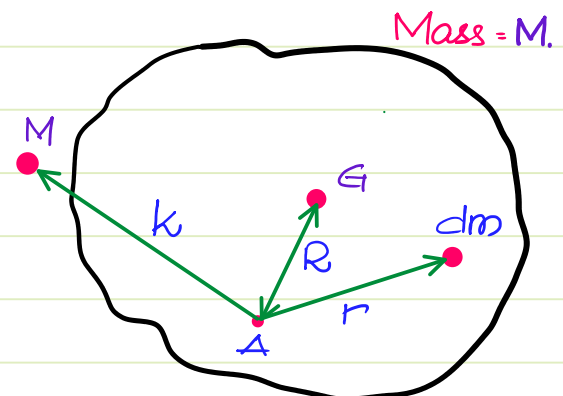


B. Rotational Inertia of Rigid Body :

$$dI_A = dm \cdot r^2 \quad \{I \text{ of } "dm" \text{ abt } A\}$$

$I_A = \int_M dm \cdot r^2$

first principle method.



Parallel Axis Th^{rm}: $I_A = I_G + MR^2$

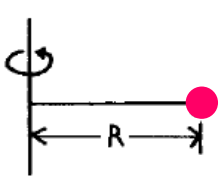
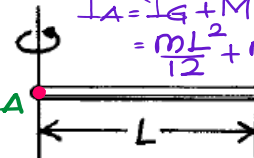
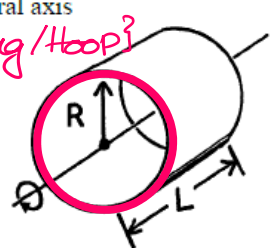
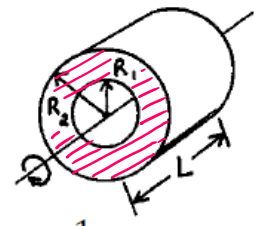
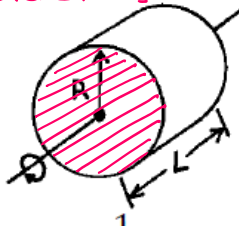
- * I_G is Mass M.O.I. abt an axis $\parallel$ to that at A & passing thru G.
- * R is the distance b/w A & G.

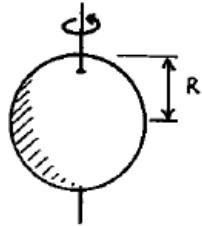
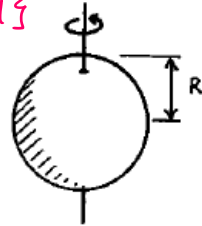
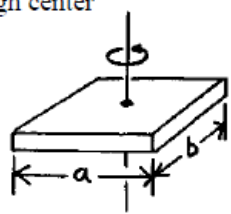
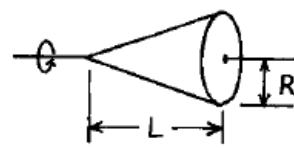
Radius of Gyration: The distance measured from the axis of rotation at which the entire mass is concentrated as a particle such that it creates the same inertia as that by the rigid body.

$$I_A = M \cdot k^2 \Rightarrow k = \sqrt{\frac{I_A}{M}}$$

If radius of gyration is given, then it is wrt centre of gravity by default.

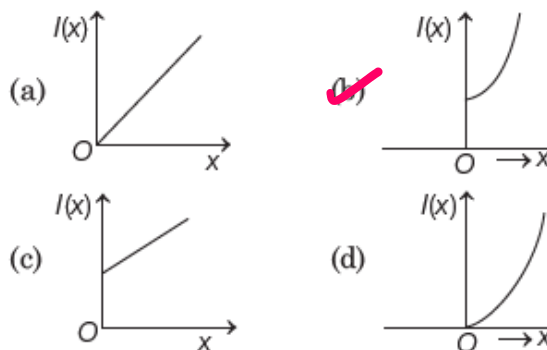
Mass M.O.I. of Rigid Bodies:

Point mass at a radius R  $I = MR^2$	Thin rod about axis through center perpendicular to length <i>Slender Rod</i>  $I = \frac{1}{12} ML^2$	Thin rod about axis through end perpendicular to length  $I_A = I_G + M\{GA\}^2$ $= \frac{ML^2}{12} + m\left(\frac{L}{2}\right)^2 = \frac{mL^2}{3}$ $I = \frac{1}{3} ML^2$
Thin-walled cylinder about central axis <i>{Ring/Hoop}</i>  $I = MR^2$	Thick-walled cylinder about central axis <i>Annular disc</i>  $I = \frac{1}{2} M(R_1^2 + R_2^2)$	Solid cylinder about central axis <i>{Solid disc}</i>  $I = \frac{1}{2} MR^2$

Solid sphere about center  $I = \frac{2}{5} MR^2$	Thin hollow sphere about center <i>{shell}</i>  $I = \frac{2}{3} MR^2$
Slab about perpendicular axis through center  $I = \frac{1}{12} M(a^2 + b^2)$	Cone about central axis  $I = \frac{3}{10} MR^2$

IM.

The moment of inertia of a solid sphere, about an axis parallel to its diameter and at a distance of x from it, is ' $I(x)$ '. Which one of the graphs represents the variation of $I(x)$ with x correctly?



$$I(x) \text{ at } x=0 \\ \left\{ I = \frac{2}{5} MR^2 \right\} \neq 0$$

$$I(x) \text{ at } x \neq 0 \\ \text{Parallel Axis Thm:} \\ I(x) = \frac{2}{5} MR^2 + Mx^2 \\ = I + Mx^2$$

Newton's 2nd Law of Motion in FAR: $\because \vec{a} = 0 \therefore \Sigma \vec{F} = 0$

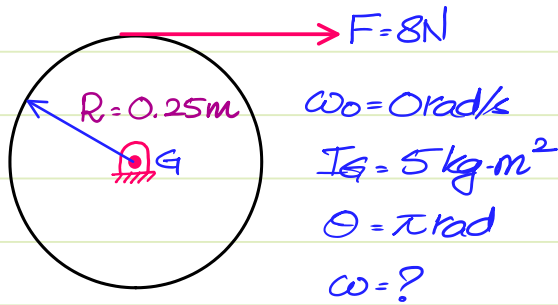
$\because$ Rotation is present, $\therefore \alpha$ may not be equal to zero.

$$\Sigma \vec{M} \neq 0$$

$$\Sigma M = I\alpha$$

Note: I must be considered about the axis about which Moment is considered.

IM.



Kinetics:

$$\Sigma M_G = I_G \alpha$$

$$F(R) = I_G \alpha$$

$$\alpha = 0.4 \text{ rad/s}^2$$

Kinematics: $\omega^2 - \omega_0^2 = 2\alpha\theta$

$$\omega = \sqrt{2 \times 0.4 \times \pi}$$

$$\Rightarrow \omega = 1.58 \text{ rad/s}$$

12. A disk with a rotational inertia of $5.0 \text{ kg}\cdot\text{m}^2$ and a radius of 0.25 m rotates on a frictionless fixed axis perpendicular to the disk and through its center. A force of 8.0 N is applied tangentially to the rim. If the disk starts at rest, then after it has turned through half a revolution its angular velocity is
- (a) 0.57 rad/s (b) 0.64 rad/s
 (c) 0.80 rad/s (d) 1.6 rad/s

IM:

$$T = 60 \text{ N}\cdot\text{m}$$

$$T = I \alpha \Rightarrow \alpha = \frac{60 \times 12}{40 \times 2^2}$$

$$\Rightarrow \alpha = 4.5 \text{ rad/s}^2$$

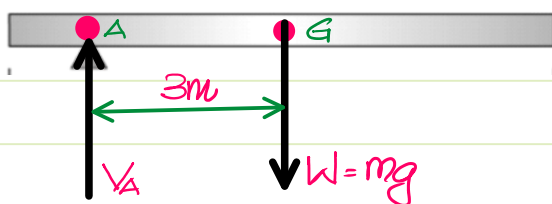
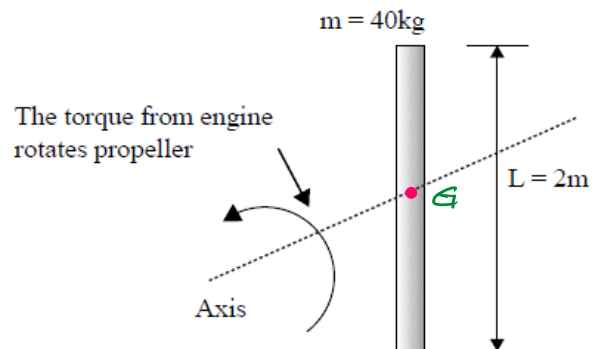
$$\omega = \frac{2\pi N}{60} = \frac{20\pi}{3}$$

$$\omega = \omega_0 + \alpha t \Rightarrow t = \omega / \alpha$$

$$t = \frac{20\pi}{3 \times 4.5}$$

$$t = 4.65 \text{ s}$$

13. The engine in a small airplane is specified to have a torque of $60 \text{ N}\cdot\text{m}$. This engine drives a 2 m long 40 kg propeller. On start up, the time taken by the propeller to reach 200 rpm is _____.

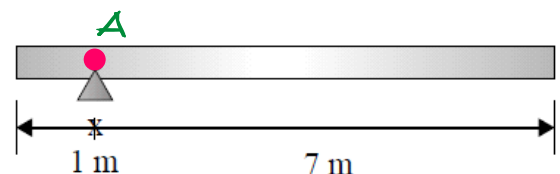


$$\Sigma M_A = mg\{3\}$$

$$I_A = \frac{mL^2}{12} + m\{3\}^2$$

$$\alpha = \frac{\Sigma M_A}{I_A} \Rightarrow \alpha = 2.053 \text{ rad/s}^2$$

10. A uniform slender rod (8 m length and 3 kg mass) rotates in a vertical plane about a horizontal axis 1 m from its end as shown in the figure. The magnitude of the angular acceleration (in rad/s^2) of the rod at the position shown is _____. (GATE- 2014)



Newton's 2nd Law of Motion in GPM. {General Plane Motion}

∴ the body observes both translation & rotation

$$\therefore \sum \vec{F} = m\vec{a} \quad \{\vec{a} \text{ of C.O.G. of the body}\}$$

$$\sum \vec{M} = I \cdot \alpha$$

If D'Alembert's Principle is used :

Inertia force = ma opposite to the dirⁿ of $\vec{a}$

Inertia Moment = $I\alpha$ opposite to the dirⁿ of α

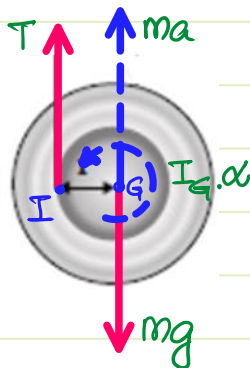
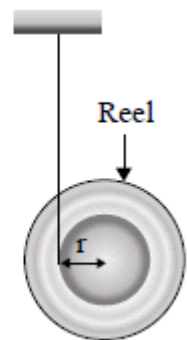
V. Imp Note : Relationship b/w a & α .

$$a = r \cdot \alpha$$

Where r is the distance from I-Centre of the body to G.

A reel of mass " m " and radius of gyration " k " is rolling down smoothly from rest with one end of the thread wound on it held in the ceiling as depicted in the figure below. Considering the thickness of the thread and its mass negligible in comparison with the radius " r " of the reel and its mass " m ", the linear acceleration of the reel is (GATE-2002)

- (a) $\frac{gr^2}{(r^2 + k^2)}$
 (b) $\frac{gk^2}{(r^2 + k^2)}$
 (c) $\frac{grk}{(r^2 + k^2)}$
 (d) $\frac{mgr^2}{(r^2 + k^2)}$



$$\sum F_y = 0 : T + ma = mg$$

$$\Rightarrow T = mg - ma$$

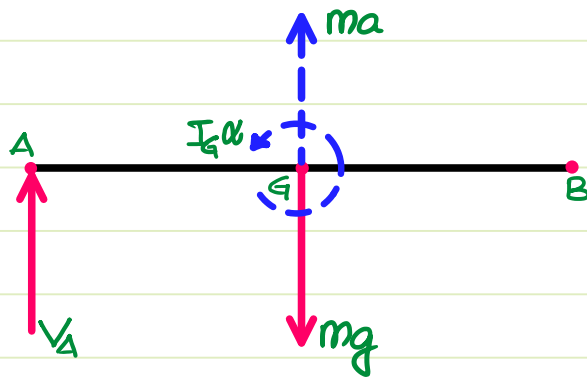
$$\sum M_A = 0 : -mg\{r\} + ma\{r\} + I_G \cdot \alpha = 0$$

$$I_G = mk^2 ; \alpha = \frac{a}{r}$$

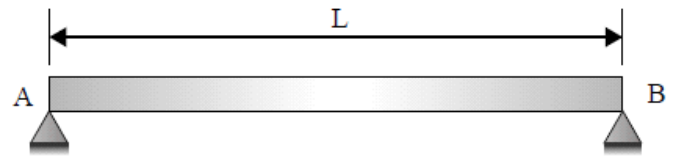
$$\Rightarrow \cancel{mar} + mk^2 \frac{a}{r} = \cancel{mgr}$$

$$\Rightarrow a\{r^2 + k^2\} = gr^2$$

$$\Rightarrow a = g \cdot \left\{ \frac{r^2}{r^2 + k^2} \right\}$$



11. As shown in the figure below a slender bar of weight W and length L is supported at both ends. The reaction at A the instant after the support at B is suddenly removed is (GATE - 13)



Pt A is the I-Centre $\therefore \alpha = a/r_{AG} = a/L/2 = \frac{2a}{L}$
 $I_G = \frac{mL^2}{12}$

$$\sum F_y = 0: V_A - mg + ma = 0$$

$$\Rightarrow V_A = mg - ma$$

$$\sum M_A = 0: -mg\left\{\frac{L}{2}\right\} + ma\left\{\frac{L}{2}\right\} + I_G \alpha = 0$$

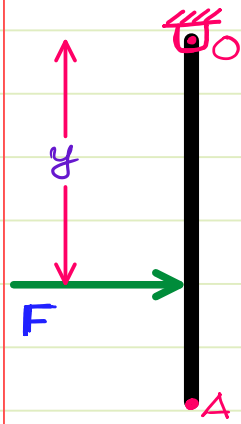
$$\Rightarrow \cancel{\frac{mgL}{2}} = \cancel{\frac{maL}{2}} + \cancel{\frac{mL^2}{12}} \cdot \cancel{\frac{2a}{L}}$$

$$\Rightarrow g/2 = 2a/3$$

$$\Rightarrow a = \frac{3g}{4}$$

$$\Rightarrow V_A = mg - m\left\{\frac{3g}{4}\right\}$$

$$\therefore V_A = \frac{mg}{4} = \frac{W}{4}$$



If the weight of the slender rod is W & the length is L , then at what distance " y " must the force be applied so that the horizontal reaction at hinge is zero.

$$\sum F_x = 0:$$

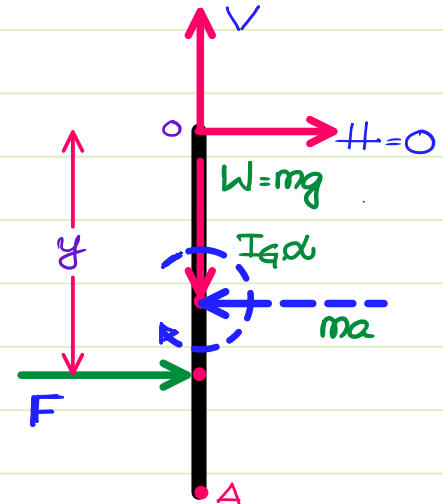
$$\Rightarrow F = ma$$

$$\sum M_O = 0$$

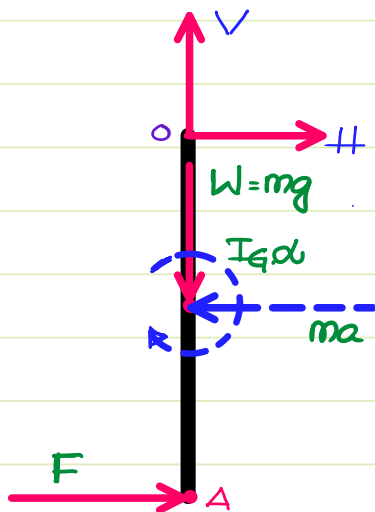
$$F\{y\} - ma\{L/2\} - I_G \alpha = 0$$

$$may = \frac{maL}{2} + \frac{mL^2}{12} \cdot \frac{2a}{L}$$

$$y = 2L/3$$



* Find the hinge reacⁿ & accⁿ of the rod if force is applied at "A"



$$\sum F_x = 0$$

$$\Rightarrow F + H = ma$$

$$\sum M_O = 0$$

$$F\{L\} - ma\{L/2\} - I_G \alpha = 0$$

$$FL = ma(L/2) + \frac{mL^2}{12} \cdot \frac{2a}{L}$$

$$\Rightarrow F = \frac{ma}{2} + \frac{ma}{6} \Rightarrow F = \frac{2ma}{3}$$

$$\therefore a = \frac{3F}{2M} \Rightarrow H = F/2$$

What would be the value of accⁿ & horizontal hinge reacⁿ if F is applied at "G"

$$a = \frac{3F}{4M} \Rightarrow H = -F/4$$

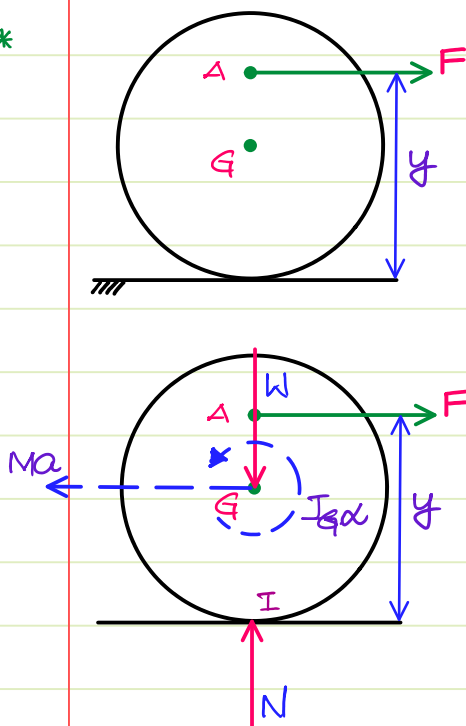
Rolling Friction

The friction acting on a body rolling on a surface is Rolling friction. There are 2 types of Rolling friction.

- A. Static Rolling friction
- B. Dynamic / Kinetic Rolling friction.

- A. Static Rolling friction: i) acts when body rolls w/o slip.
 ii) value ranges from 0 to $f_s \{ = \mu_s N \}$
 iii) It may act in the direction or opposite to the direction of applied force. {direction can be identified only after solving the dyn eq^{ns}}
- B. Kinetic Rolling friction: i) acts when body rolls with slip.
 ii) value is always equal to $f_k \{ = \mu_k N \}$
 iii) It acts in a direction opposite to the slip {relative motion}.

**



Given:

A solid sphere of mass $\{M\}$ & radius $\{R\}$.

To find:

The location "y" at which a horizontal force applied creates 0 friction while having rolling w/o slip.

$$\sum M_I = 0$$

$$-Fy + MaR + \frac{2}{5}MR^2 \frac{a}{R} = 0$$

$$y = \frac{Ma}{F} \left\{ \frac{7R}{5} \right\}$$

$$\sum F_x = 0$$

$$F = Ma$$

$$\therefore y = \frac{7R}{5}$$

Slender Rod: $y = \frac{2L}{3}; H=0; a = \frac{F}{M}$ | $y > \frac{2L}{3} H > 0 a > \frac{F}{M}$ | $y < \frac{2L}{3} H < 0 a < \frac{F}{M}$

Solid Sphere: $y = \frac{7R}{5}; f=0; a = \frac{F}{M}$ | $y > \frac{7R}{5} f > 0 a > \frac{F}{M}$ | $y < \frac{7R}{5} f < 0 a < \frac{F}{M}$

Home Work :

find the acceleration & friction force in the above problem if force is applied at

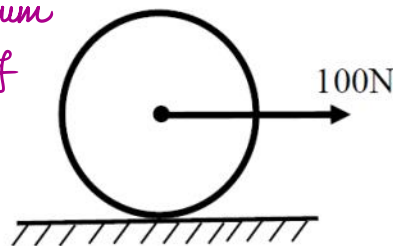
a: $y = 2R$

b: $y = R$

2M.

A force of 100 N is applied to the centre of a circular disc, of mass 10 kg and radius 1 m, resting on a floor as shown in the figure. If the disc rolls without slipping on the floor, the linear acceleration (in m/s^2) of the centre of the disc is _____ (correct to two decimal places).

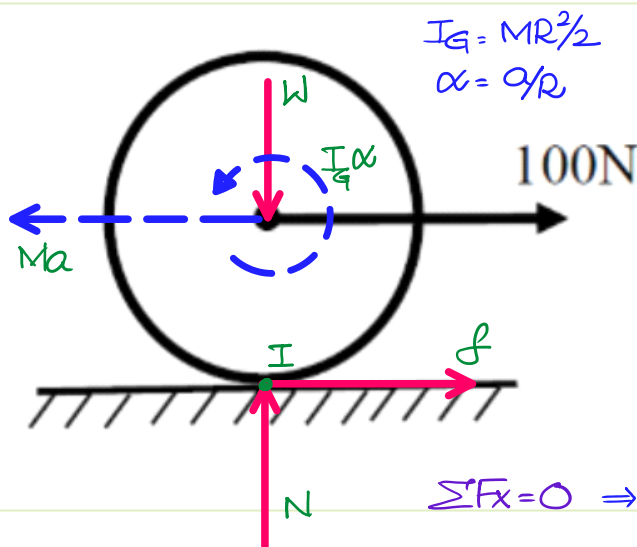
* Also find the minimum value of co-eff of friction



If body rolls w/o slip on a fixed surface:

* the pt. of contact is I Centre.

* $a = R\alpha \Rightarrow \alpha = \frac{a}{R}$



$I_G = MR^2/2$
 $\alpha = a/R$

$\sum M_I = 0$

$-100(R) + Ma(R) + \frac{MR^2}{2} \cdot \frac{a}{R} = 0$

$\Rightarrow \frac{3Ma}{2} = 100 \Rightarrow a = \frac{200}{3M}$

$\therefore a = \frac{20}{3} m/s^2 \approx 6.67 m/s^2$

$\sum F_x = 0 \Rightarrow 100 + f = Ma$

$\Rightarrow f = 10 \left\{ \frac{20}{3} \right\} - 100$

$\therefore f = \frac{100}{3} N$

At Minimum coeff of static friction:

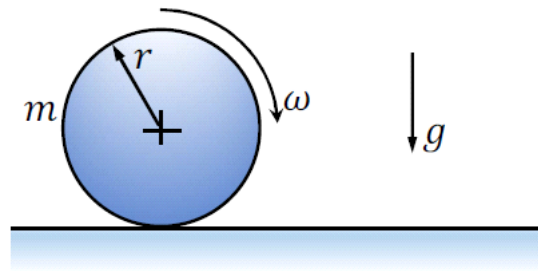
$f = f_s = \mu_s N$

$\therefore \mu_s = \frac{100/3}{100}$

$\therefore \{\mu_s\}_{min} = \frac{1}{3}$

2M, ME

A cylindrical disc of mass $m = 1$ kg and radius $r = 0.15$ m was spinning at $\omega = 5$ rad/s when it was placed on a flat horizontal surface and released (refer to the figure). Gravity g acts vertically downwards as shown in the figure. The coefficient of friction between the disc and the surface is finite and positive. Disregarding any other dissipation except that due to friction between the disc and the surface, the horizontal velocity of the center of the disc, when it starts rolling without slipping, will be _____ m/s (round off to 2 decimal places).



When the disc is released it is observing **FAR**.

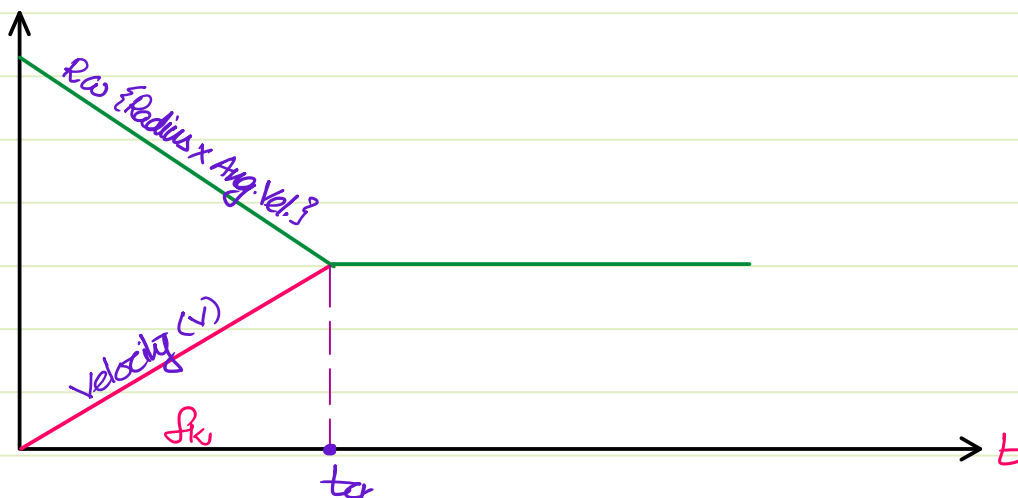
The point on the disc @ contact with the surface is slipping in the backward dirⁿ $\therefore$ f_k acts in forward dirⁿ.

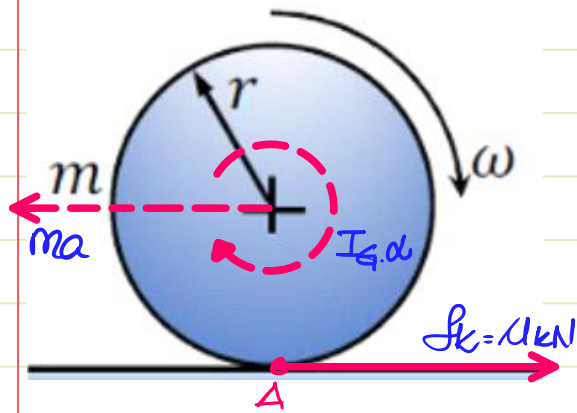
What is $f_k = \mu_k N$ doing?

* $\therefore f_k$ acts forward $\therefore a$ is forward, $\therefore$ the disc develops a forward velocity.

* The torque by f_k is anticlockwise, while ω is clockwise, $\therefore \omega$ decreases

At $t = t_{cr}$ $V = R\omega$ {Rolling w/o slip}





$\therefore$ Rolling with slip is taking place
 $\therefore$ We can't write $a = R\alpha$.

Step 1: Kinetics { find relation b/w a & α }

$$\sum M_A = 0$$

$$MaR - I_G \alpha = 0$$

$$\Rightarrow MaR = \frac{MR^2}{2} \alpha$$

$$\Rightarrow \boxed{a = \frac{R\alpha}{2}}$$

Step 2: Kinematics :

$$v = v_0 + at$$

$$v = \frac{R\alpha}{2} \cdot t \Rightarrow \boxed{t = \frac{2v}{R\alpha}}$$

$$\omega = \omega_0 + \alpha t$$

$$\omega = 5 - \cancel{\alpha} \frac{2v}{R\alpha} \Rightarrow \omega = 5 - \frac{2v}{R}$$

At Rolling w/o slip: $v = R\omega$

$$v = R \left\{ 5 - \frac{2v}{R} \right\}$$

$$\Rightarrow v = 5R - 2v \Rightarrow 3v = 5R$$

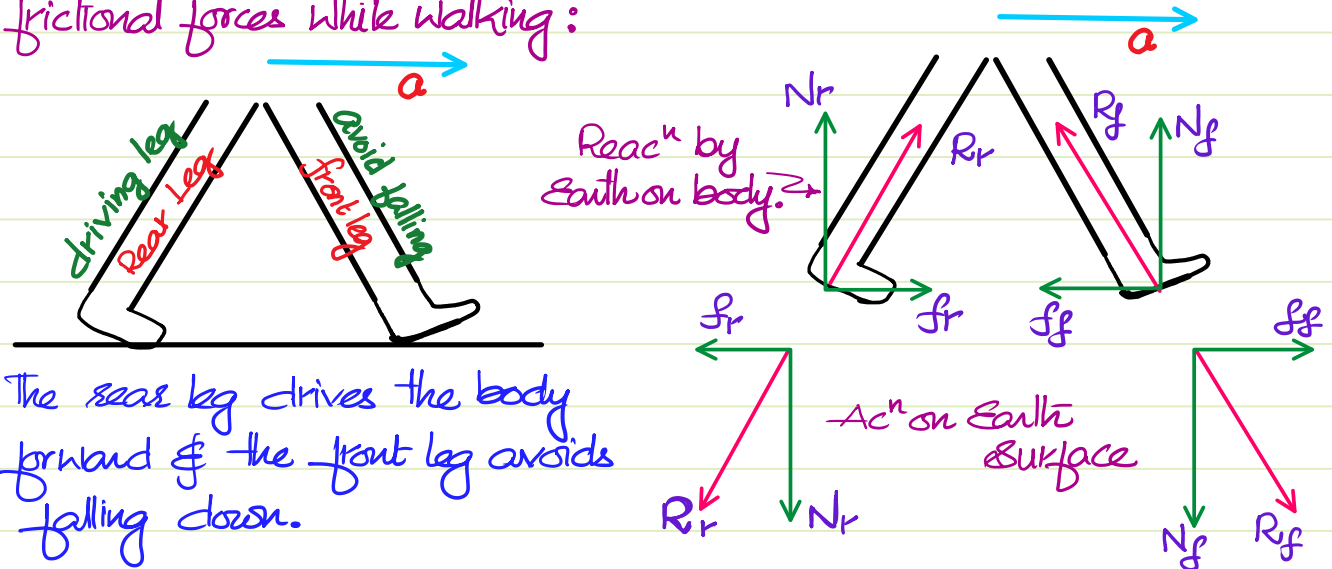
$$\Rightarrow v = \frac{5R}{3} = \frac{5 \times 0.15}{3}$$

$$\boxed{v = 0.25 \text{ m/s}}$$

Note: In the absence of a driving force, if a body rolls w/o slip, then there is no friction.

* Vehicle Friction : The frictional force b/w wheel of a vehicle & road is known as vehicle friction.

frictional forces while walking :



Friction on the rear leg {driving leg} is in forward direction
Friction on the front leg {avoiding to fall} is in backward direction

$$\sum F_x = ma \Rightarrow f_r - f_f = m \cdot a \Rightarrow a = \frac{f_r - f_f}{m}$$

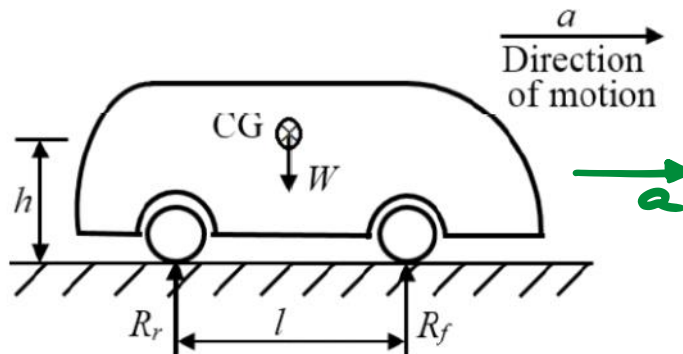
* Friction in Bicycle :



driving wheel applies forces on Earth such that it gets friction in the direction it wants the acceleration.

Always assume the driven wheel initially is frictionless, such that it has pure translation. Then introduce friction so as to create rolling w/o slip.

A car having weight W is moving in the direction as shown in the figure. The center of gravity (CG) of the car is located at height h from the ground, midway between the front and rear wheels. The distance between the front and rear wheels is l . The acceleration of the car is a , and acceleration due to gravity is g . The reactions on the front wheels (R_f) and rear wheels (R_r) are given by

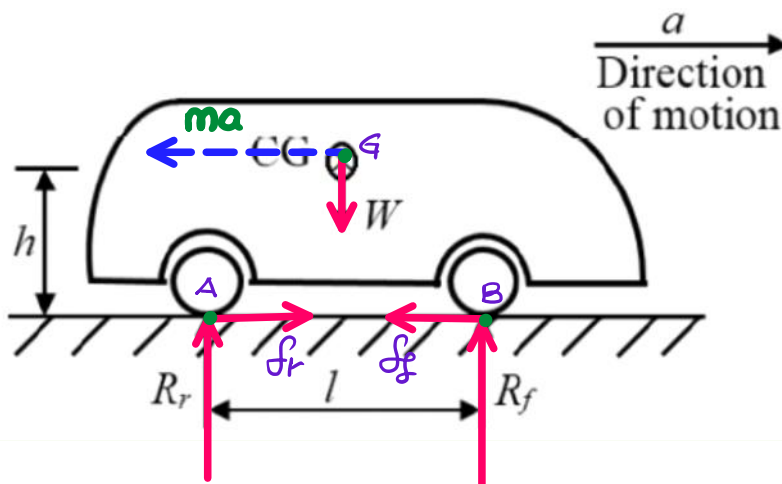


(A) $R_f = R_r = \frac{W}{2} - \frac{W}{g} \left(\frac{h}{l} \right) a$

(B) $R_f = \frac{W}{2} + \frac{W}{g} \left(\frac{h}{l} \right) a$; $R_r = \frac{W}{2} - \frac{W}{g} \left(\frac{h}{l} \right) a$

✓ (C) $R_f = \frac{W}{2} - \frac{W}{g} \left(\frac{h}{l} \right) a$; $R_r = \frac{W}{2} + \frac{W}{g} \left(\frac{h}{l} \right) a$

(D) $R_f = R_r = \frac{W}{2} + \frac{W}{g} \left(\frac{h}{l} \right) a$



$$\sum F_y = 0$$

$$\Rightarrow R_r + R_f = W$$

$$\sum M_A = 0$$

$$-W\left(\frac{l}{2}\right) + R_f(l) + ma\{h\} = 0$$

$$R_f = \frac{W}{2} - \frac{W}{g} a \left\{ \frac{h}{l} \right\}$$

$$R_r = \frac{W}{2} + \frac{W}{g} a \left\{ \frac{h}{l} \right\}$$

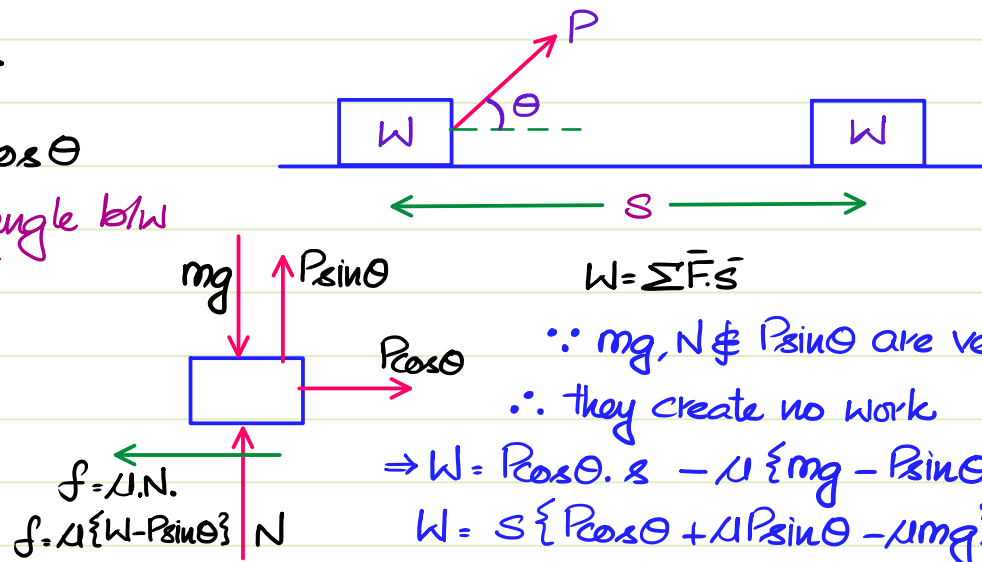
4.2. Work - Energy ^{Th^{con}}

Work: The scalar product of force & displacement is known as Work.

$$W = \vec{F} \cdot \vec{s}$$

$$= Fs \cos \theta$$

θ is the angle b/w $\vec{F}$ & $\vec{s}$.



Work done by a Moment {or} Torque: $W = \vec{T} \cdot \vec{\theta}$

Energy: Capacity of a body to do work is known as Energy.
 $\therefore$ Work & energy have same units & dimensions.

S.I.: $N \cdot m = J$ Dim: ML^2T^{-2}

The subject of Engg. Mech understands only 2 energies.

1. Kinetic Energy {K}
2. Potential Energy {U}

* Kinetic Energy {K}: Energy of a body due to its motion wrt. time.
 $\therefore$ Kinetic energy depends on Velocity.

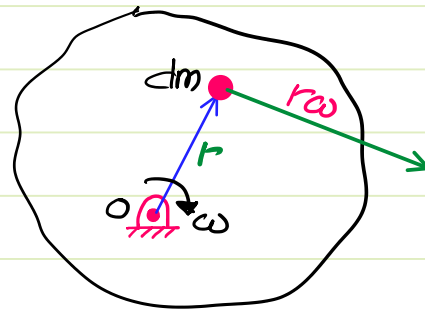
A. Translational Kinetic Energy:

A body of Mass {M} has a velocity {V}

$K_T = \frac{1}{2} MV^2$

B. Rotational Kinetic Energy

$$\begin{aligned} dK &= \frac{1}{2} dm v^2 \\ &= \frac{1}{2} dm r^2 \omega^2 \\ K_R &= \frac{1}{2} \omega^2 \int dm r^2 \end{aligned}$$

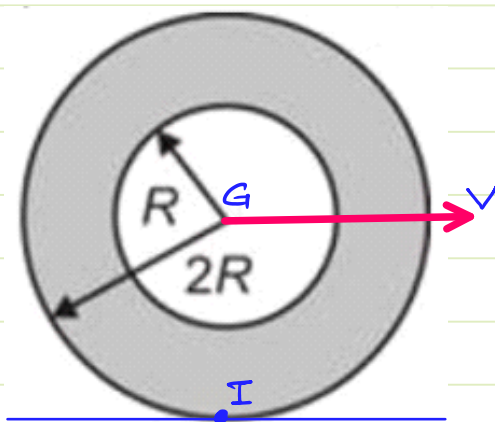


$$\therefore K_R = \frac{1}{2} I_O \omega^2 \quad \text{I must be always abt the centre of FAR}$$

C. Kinetic Energy of A Body In General Plane Motion :

$$K = K_T + K_R \Rightarrow K = \frac{1}{2} M V^2 + \frac{1}{2} I_G \omega^2 \quad \text{I: abt centre of Gravity}$$

$$\therefore \text{GPM} = \text{FAR abt I centre : } K = \frac{1}{2} I_I \omega^2 \quad \text{I: abt I centre.}$$



An annular disc has a mass m , inner radius R and outer radius $2R$. The disc rolls on a flat surface without slipping. If the velocity of the center of mass is v , the kinetic energy of the disc is

(GATE-2014)

(a) $\frac{9mv^2}{16}$

(b) $\frac{11mv^2}{16}$

☒ (c) $\frac{13mv^2}{16}$

(d) $\frac{15mv^2}{16}$

$$\begin{aligned} K &= K_T + K_R \\ &= \frac{1}{2} m v^2 + \frac{1}{2} I_G \omega^2 \end{aligned}$$

$$\begin{aligned} I_G &= \frac{m}{2} \{ (2R)^2 + R^2 \} \\ &= \frac{5mR^2}{2} \end{aligned}$$

$$\omega = v/2R$$

$$K = \frac{1}{2} m v^2 + \frac{1}{2} \left\{ \frac{5mR^2}{2} \right\} \frac{v^2}{4R^2} = \frac{13}{16} m v^2$$

Alternate Method: $K = \frac{1}{2} I_I \omega^2$

$$I_I = I_G + m \{ 2R \}^2$$

$$= \frac{5}{2} m R^2 + 4 m R^2 = \frac{13}{2} m R^2$$

$$K = \frac{1}{2} \left\{ \frac{13}{2} m R^2 \right\} \left\{ \frac{v}{2R} \right\}^2$$

$$K = \frac{13}{16} m v^2$$

* Potential Energy $\{U\}$: The energy stored in a body due to work done on it against a conservative field of force.

A. Gravitational Potential Energy $\{U_g\}$: $U_g = mgh$

* g is a constant.

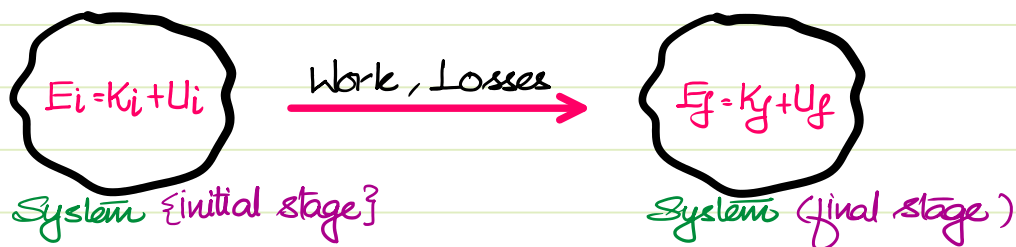
* h is +ve upward & meas. from ref. datum

B. Spring Potential Energy $\{U_s\}$: $U_s = \frac{1}{2} kx^2$

* Spring const $\{k\}$ is constant

* x is deformed length.

* Work-Energy Th^{rm}:



The relationship b/w the energies of a system & the work + loss interactions with surroundings is Work-Energy Th^{rm}.

$$E_i + W - L = E_f \Rightarrow \{E_f - E_i\} = W - L$$

$\Delta E = W - L$

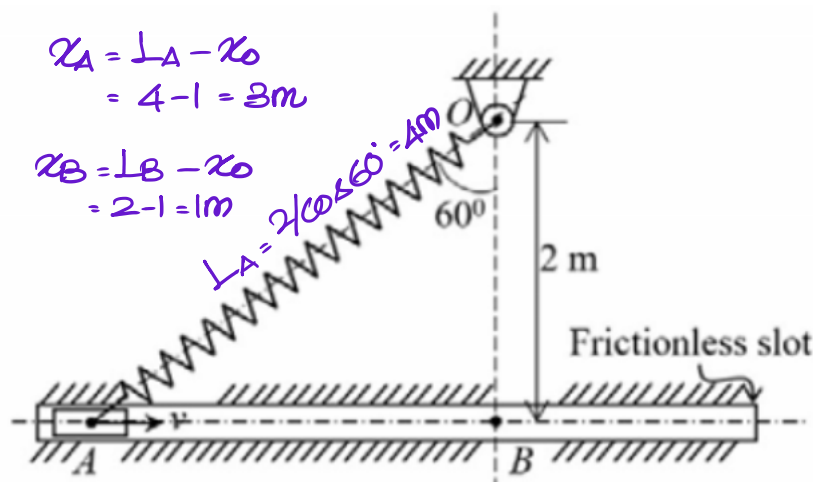
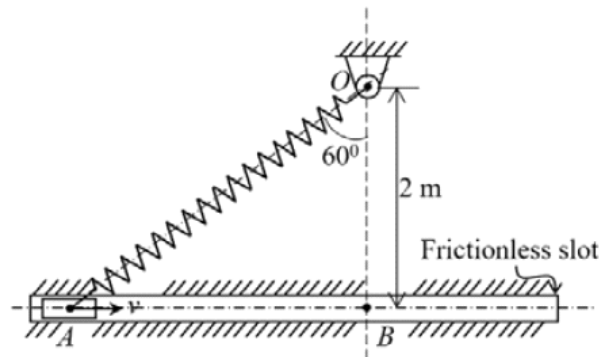
ΔE : Change in energy of the system $\{E_f - E_i\}$

W : Work done on the system

L : Losses observed by the system

Note: Prefer Work-Energy Th^{rm} when finding accⁿ of the system is difficult.

A slider moving in a frictionless slot is connected with a linear spring OA as shown in the figure. The following is known: stiffness of the spring = 2 kN/m, mass of the slider = 10 kg, and the unstretched length of the spring = 1 m. If the slider is released from rest at A the magnitude of its velocity (in m/s) when it passes through point B is _____ (rounded off to the nearest integer)



As the slider moves from A to B , its accⁿ is variable.

$\therefore$ Work-Energy th^m is pref.

$$E_B - E_A = W - L$$

$$E_B = E_A$$

$$K_B + U_B = K_A + U_A$$

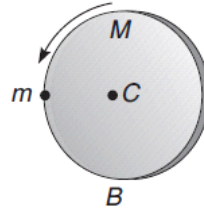
$$K_B = \frac{1}{2} m v_B^2 = \frac{1}{2} k \{x_A^2 - x_B^2\}$$

$$\Rightarrow v_B = \sqrt{\frac{k}{m} \{x_A^2 - x_B^2\}}$$

$$v_B = \sqrt{\frac{2000}{10} \{3^2 - 1^2\}}$$

$$v_B = 40 \text{ m/s}$$

A uniform disc of mass M and radius R is pivoted about the horizontal axis through its centre C . A point mass m is glued to the disc at its rim, as shown in figure. If the system is released from rest, find the angular velocity of the disc when m reaches the bottom point B .



Assume the system is frictionless $E_B = E_A$

If the ref. datum is taken at B.
Then $U_B = 0$ & $U_A = mg\{2R\}$

$$K_A = 0 ; K_B = \frac{1}{2}\{I_G M\}\omega^2 + \frac{1}{2}mv^2$$

$$K_B = \frac{1}{2}\left\{\frac{1}{2}MR^2\right\}\omega^2 + \frac{1}{2}m(R\omega)^2$$

$$K_B = \frac{1}{2}\omega^2\left\{\frac{MR^2}{2} + mR^2\right\}$$

$$\cancel{K_B + U_B} = \cancel{K_A} + U_A$$

$$\frac{1}{2}\omega^2 R^2 \left\{ \frac{M}{2} + m \right\} = 2mgR$$

$$\Rightarrow \omega = 2 \sqrt{\frac{m \cdot g}{R \left\{ \frac{M}{2} + m \right\}}} = 2 \sqrt{\frac{2mg}{R \{M + 2m\}}}$$

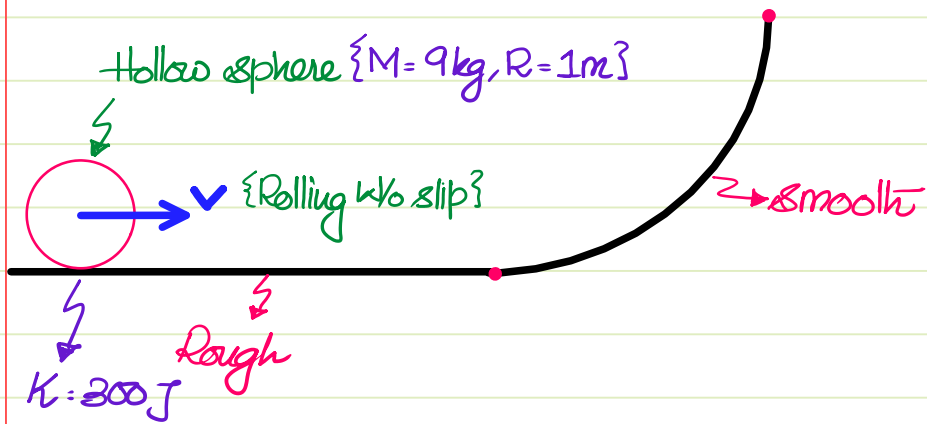
A body rolls down an inclined plane without slipping. The kinetic energy of rotation is 50% of its translational kinetic energy. The body is

- (a) Solid sphere
- (b) Hollow cylinder
- ☒ (c) Solid cylinder
- (d) Ring

$$K = \frac{1}{2}mv^2 + \frac{1}{2}I_G \omega^2$$

$$\text{Given: } \frac{1}{2}I_G \omega^2 = \frac{1}{2} \left\{ \frac{1}{2}mv^2 \right\}$$

$$\Rightarrow I_G = \frac{m}{2} \left\{ \frac{v^2}{\omega^2} \right\} = \frac{mR^2}{2}$$



Find the Max height
 The hollow sphere
 can climb over the
 smooth surface.

When the body climbs over smooth surface, its rotational kinetic energy is unaffected $\{f=0\}$

$\Rightarrow$ only translational KE $\rightarrow U_g$

$$K = \frac{1}{2} I_{\text{I}} \cdot \omega^2 = \frac{1}{2} I_{\text{I}} \cdot \left\{ \frac{V}{R} \right\}^2$$

$$300 = \frac{1}{2} \times \frac{5}{3} m R^2 \cdot \frac{V^2}{R^2}$$

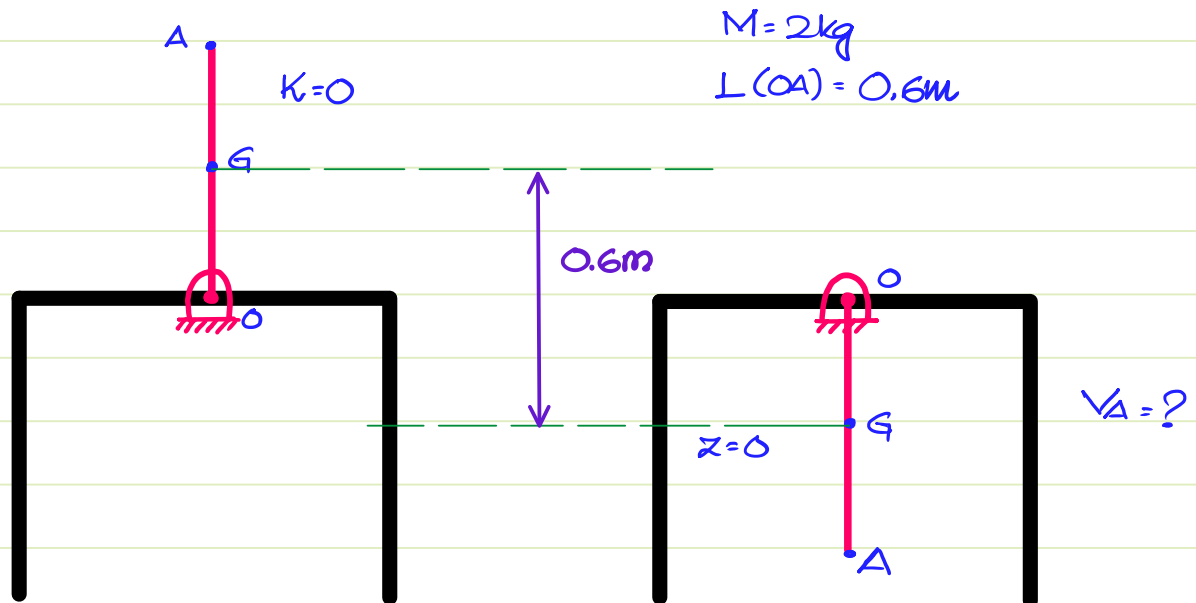
$$\Rightarrow V = \sqrt{40} \text{ m/s}$$

$$K_T = U_g \Rightarrow \frac{1}{2} m v^2 = m g h$$

$$\Rightarrow h = \frac{v^2}{2g} = \frac{40}{2 \times 10}$$

$$\Rightarrow h = 2\text{ m}$$

A **2 kg** steel rod of length **0.6 m** is clamped on a table vertically at its lower end and is free to rotate in vertical plane. The upper end is pushed so that the rod falls under gravity, Ignoring the friction due to clamping at its lower end, the speed of the free end of rod when it passes through its lowest position is (Take $g = 10 \text{ ms}^{-2}$)



$$E_f - E_i = \cancel{W} - \cancel{L}$$

$$\Rightarrow E_f = E_i$$

$$K_f + \cancel{U_f} = \cancel{K_i} + U_i$$

$$\frac{1}{2} I_o \cdot \omega^2 = mg \{0.6\}$$

$$\Rightarrow \frac{1}{2} \frac{mL^2}{3} \cdot \omega^2 = mg (0.6)$$

$$\Rightarrow \omega = \sqrt{\frac{6 \times 0.6g}{L^2}} \Rightarrow \omega = 10 \text{ rad/s}$$

$$V_A = r_{Ao} \cdot \omega$$

$$= 0.6 \times 10$$

$$V_A = 6 \text{ m/s}$$

4.3. Lagrangian Method.

This is a scalar method useful in finding the eq^{ns} of motion in complex systems involving Multiple degrees of freedom. It doesn't require drawing FBD & writing Newton's 2nd Law of Motion.

Lagrangian Function $\{L\}$: It is the difference b/w the kinetic energy $\{K\}$ & potential energy $\{U\}$ of the system at any arbitrary time instant.

$$L = K - U \quad \text{SI: J = N.m}$$

If a system has "n" degrees of freedom, then

$$L = f(q_1, q_2, \dots, q_n, \dot{q}_1, \dot{q}_2, \dots, \dot{q}_n, t)$$

↓
Positions in n dof.

↓
Velocities in n dof.

Eq^{ns} of Motion:

$$\frac{d}{dt} \left\{ \frac{\partial L}{\partial \dot{q}} \right\} - \frac{\partial L}{\partial q} = 0$$

Generalised
eqⁿ of Motion.

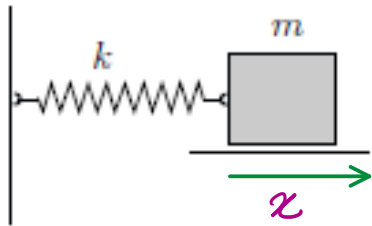
If the system has "n" dof, then there are "n" eq^{ns} of Motion.

$$\frac{d}{dt} \left\{ \frac{\partial L}{\partial \dot{q}_1} \right\} - \frac{\partial L}{\partial q_1} = 0$$

$$\frac{d}{dt} \left\{ \frac{\partial L}{\partial \dot{q}_2} \right\} - \frac{\partial L}{\partial q_2} = 0$$

⋮

$$\frac{d}{dt} \left\{ \frac{\partial L}{\partial \dot{q}_n} \right\} - \frac{\partial L}{\partial q_n} = 0$$



Find the eqⁿ of Motion for the spring-mass system using Lagrangian Method.

Given system has 1 DOF {x}

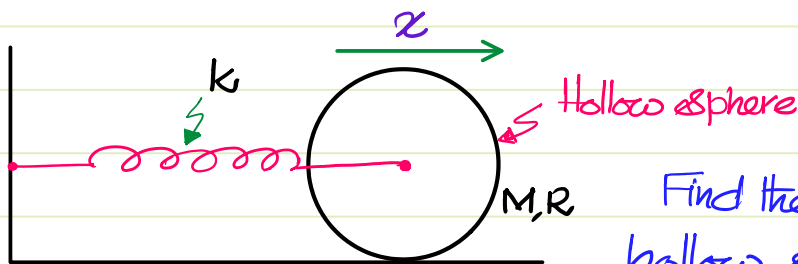
$$K = \frac{1}{2}mv^2 = \frac{1}{2}m\dot{x}^2 ; U = \frac{1}{2}kx^2 \quad \therefore L = \frac{1}{2}m\dot{x}^2 - \frac{1}{2}kx^2$$

$$\frac{\partial L}{\partial \dot{x}} = m\dot{x} \Rightarrow \frac{d}{dt} \left\{ \frac{\partial L}{\partial \dot{x}} \right\} = m\ddot{x}$$

$$\frac{\partial L}{\partial x} = -kx$$

Eqⁿ of Motion: $\frac{d}{dt} \left\{ \frac{\partial L}{\partial \dot{x}} \right\} - \frac{\partial L}{\partial x} = 0$

$$\Rightarrow m\ddot{x} + kx = 0 \Rightarrow \ddot{x} = -\frac{k}{m} \cdot x$$



Find the eqⁿ of Motion if the hollow sphere rolls w/o slip.

$$U = \frac{1}{2}kx^2$$

$$K = \frac{1}{2}I_I \cdot \omega^2$$

$$K = \frac{1}{2} \left\{ \frac{5}{3}MR^2 \right\} \frac{v^2}{R^2}$$

$$\Rightarrow K = \frac{5}{6}M\dot{x}^2$$

$$\frac{\partial L}{\partial \dot{x}} = \frac{5}{3}M\dot{x} \Rightarrow \frac{d}{dt} \left\{ \frac{\partial L}{\partial \dot{x}} \right\} = \frac{5}{3}M\ddot{x}$$

$$\frac{\partial L}{\partial x} = -kx$$

Eqⁿ of Motion: $\frac{d}{dt} \left\{ \frac{\partial L}{\partial \dot{x}} \right\} - \frac{\partial L}{\partial x} = 0$

$$L = \frac{5}{6}M\dot{x}^2 - \frac{1}{2}kx^2$$

$$\Rightarrow \frac{5}{3}M\ddot{x} + kx = 0$$

$$\ddot{x} = -\frac{3k}{5M} \cdot x$$